

PAUL OFFEI

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RESEARCH INTERESTS

My research interest is in Machine Learning for natural language & visual understanding and generation with emphasis on Multimodal & Multilingual LLM Safety and Alignment. My focus is developing algorithmic solution or approach to address the risk posed by Multilingual Large Language Models (MLLM) by highlighting open-ended questions on Value Misalignment, Robustness to Attack, Misuse, Autonomous AI Risks, Agent Safety, Interpretability for MLLM Safety, Strategies to MLLM Safety in Practice, AI Governance and Policy for MLLM Safety.

EDUCATION

University of Ghana Accra, Ghana
MSc in Computer Science Jun,2023-Aug,2024

University of Ghana Accra, Ghana
BA in Economics Sep,2015-May,2019

EXPERIENCE

Nduom School Of Business and Technology Cape Coast, Ghana
Teaching Assistant Nov,2024-date

- Joined the instructional team in organizing Artificial Intelligence course with practical emphasis on large language models and foundational models and Data Mining course with Practical focus on supervised & unsupervised machine learning techniques and also assisted in designing instructional materials for Systems Analysis & Design and Data Structures & Algorithms class.
- Partnered with the school's software development team to architect and deploy a proprietary internal AI compliance software that aligns with university's formal AI Regulation and Academic Integrity Policy, providing faculty with automated capabilities to detect academic integrity policy violations and verify the authenticity of student project work.
- Designed and delivered comprehensive, campus-wide professional development workshops for students, tutors, and faculty on Reassessing Academic Integrity in the Age of AI to drive the adoption of ethical guidelines, protect curriculum standards and responsible use of AI for academic work.

ResolutionAI Accra, Ghana
Pre-doctoral Researcher May-Aug,2025

- Synthesized papers around LLMs and AI Agents using prompt engineering for filtering to classify them into two categories; relevant and not relevant based on their title and abstract and manually evaluated 30% of the papers with 99% accuracy, with all errors being false negatives using open source models.

- Performed experiment and reverse engineering of LLM360 models to understand the underlying mechanisms of LLMs.

LG AI Research, collaboration with DataMaker

Accra, Ghana

Research Assistant

Aug,2023-Aug,2024

- Involved in bilingual alignment multi-turn datasets curation for supervised fine-tuning and direct preference optimization to improve EXAONE 3.0 instruction following ability and align to human preferences and ethics.
- Collaborated with team members to craft question-answer datasets with its corresponding "Pass" or "Fail" annotation, designed to cover a wide range of unethical and insecure scenarios including hate, bias, illegal content, information hazards such as the use of private data in training for the evaluation of EXAONE 3.0 7.8B instruction-tuned model where the model produces appropriate responses 84% of the time when faced with carefully curated adversarial queries.

TexkLab

Cape Coast, Ghana

Machine Learning Engineer

Jan,2021-Dec,2023

- Spearheaded the end-to-end development of an AI-powered agritech solution, "Real Time Automatic Detection of Mosaic Virus in Crops," specifically designed to mitigate crop loss across high-value horticultural sectors including tomatoes, peppers, and beans.
- Architected the project's technical roadmap by selecting and implementing a Convolutional Neural Network (CNN) architecture, leveraging Microsoft Azure GPU cloud infrastructure for high-performance model training.
- Conducted rigorous real-time field testing using OAK-D spatial AI cameras to ensure the product met user-centered requirements for speed and accuracy in diverse environmental conditions.

Ministry of Local Government and Rural Development

Accra, Ghana

Data Analyst

Aug,2019-Dec,2020

- Developed and deployed a consumer tracking system using Excel and MS Access during the COVID-19 lockdowns to manage emergency food distribution, successfully ensuring equitable relief allocation for over 200,000 low-income households while eliminating duplication and securing clean operational data.
- Participated in the Department of Social Welfare & Community Development field assessments data collection and parental counseling initiatives to track and re-engage youth school dropped out post-lockdown, collaborating with local NGOs to transition over 1000 disengaged students into vocational skills training within Ga East Municipal Assembly District.
- Collaborated with the Statistics Department to experiment and develop statistical machine learning models, analyzing post-pandemic youth internet exploitation and cyberbullying trends, generating data-driven insights that directly informed national social welfare policies and targeted community interventions.

HONORS & AWARDS

- Phase 1 Finalists, *OpenCV AI Competition*2021

TECHNICAL SKILLS

Project Management: Microsoft Office, Slack, Teams, Jira, Team work

Programming Languages: Python, JavaScript/TypeScript, Java, Go, Rust, C/C++

Framework & Library: Jax/PyTorch/TensorFlow, Numpy/Pandas/Matplotlib, Flask/FastAPI,

Cloud Computing: Microsoft Azure: Cloud Computing Services, AWS, GCP, Docker, Kubernetes

SOFTWARE

- *ASLockSmithGarageDoor, Professional Website project* Jan-Mar,2026
- *Grain, Open source project* 2025-date
- *Service Weaver, Open source project* 2023-2024

TALKS & PRESENTATIONS

- *Understanding Large Language Model from Scratch, Academic City University* 2025

TEACHING

COURSE INSTRUCTOR

- NBIT405 Artificial Intelligence, NSBT 2024-2026
- NBIT300 Data Mining, NSBT 2024-2026
- NBIT209 Data Structures and Algorithms, NSBT 2024-2026
- NBIT208 Systems Analysis and Design, NSBT 2024-2026

GUEST LECTURES

- *Digital Literacy, Cape Coast Technical University* 2026-date